THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a rhombus with diagonals 18 cm and 15 cm is (cm^2): 1

(a) 137 **(b)** 136 **(c)** 134 **(d)** 135

2. The area of a cyclic quadrilateral with sides 8 cm, 10 cm, 14 cm and 20 cm is (cm^2): 1

(a) 146 **(b)** 143 **(c)** 145 **(d)** 144

3. The length of an arc of angle 180° in a circle of radius 35 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 112 **(b)** 111 **(c)** 109 **(d)** 110

4. A square has the same perimeter as a circle of circumference 264 cm. The side of the square (cm) is: 1

(a) 68 **(b)** 66 **(c)** 67 **(d)** 65

5. The area of a triangle with sides 10 cm, 17 cm and 21 cm is (cm^2): 1

(a) 83 **(b)** 85 **(c)** 86 **(d)** 84

6. The semi-perimeter of a triangle with sides 11 cm, 25 cm and 30 cm is (cm): 1

(a) 33 **(b)** 32 **(c)** 34 **(d)** 35

7. The area of a circle with circumference 264 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 5546 **(b)** 5544 **(c)** 5543 **(d)** 5545

8. A wheel of radius 56 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 353 **(b)** 352 **(c)** 354 **(d)** 351

9. The area of a rhombus with diagonals 24 cm and 20 cm is (cm^2): 1

(a) 239 **(b)** 242 **(c)** 240 **(d)** 241

10. The area of a cyclic quadrilateral with sides 5 cm, 14 cm, 19 cm and 20 cm is (cm^2): 1

(a) 182 **(b)** 180 **(c)** 179 **(d)** 181

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the length of an arc of 180° in a circle of radius 35 cm (take $\pi = \frac{22}{7}$). 2

12. Find the area of a triangle with sides 10 cm, 17 cm, 21 cm. 2

13. A wheel of radius 14 cm covers a distance of 8800 cm. How many revolutions does it make (take $\pi = \frac{22}{7}$)? 2

14. Find the length of an arc of 45° in a circle of radius 28 cm (take $\pi = \frac{22}{7}$). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the triangle with sides 4 cm, 13 cm and 15 cm, and the length of the altitude on the side of 15 cm. 3

16. Find the area of the quadrilateral with sides 9, 15, 16, 20 cm that is inscribed in a circle. 3

17. Find the area of a parallelogram with base 25 cm and height 6 cm, and the base of a triangle of height 12 cm having the same area. 3

OR

Find the area of the triangle with sides 10 cm, 10 cm and 12 cm, and the length of the altitude on the side of 12 cm. 3

18. Find the area of the quadrilateral with sides 7, 8, 23, 26 cm that is inscribed in a circle. 3

OR

Find the area of a parallelogram with base 20 cm and height 8 cm, and the base of a triangle of height 16 cm having the same area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A circular track has inner radius 14 m and outer radius 28 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹2 per m^2. 5

OR

A rectangular lawn 45 m $\times$ 35 m has a path 3 m wide all around it, outside the lawn. Find the area of the path. 5

20. A circular track has inner radius 42 m and outer radius 49 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹5 per m^2. 5

OR

A rectangular lawn 60 m $\times$ 40 m has a path 5 m wide all around it, outside the lawn. Find the area of the path. 5

END OF QUESTION PAPER — Total: 40 Marks

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